

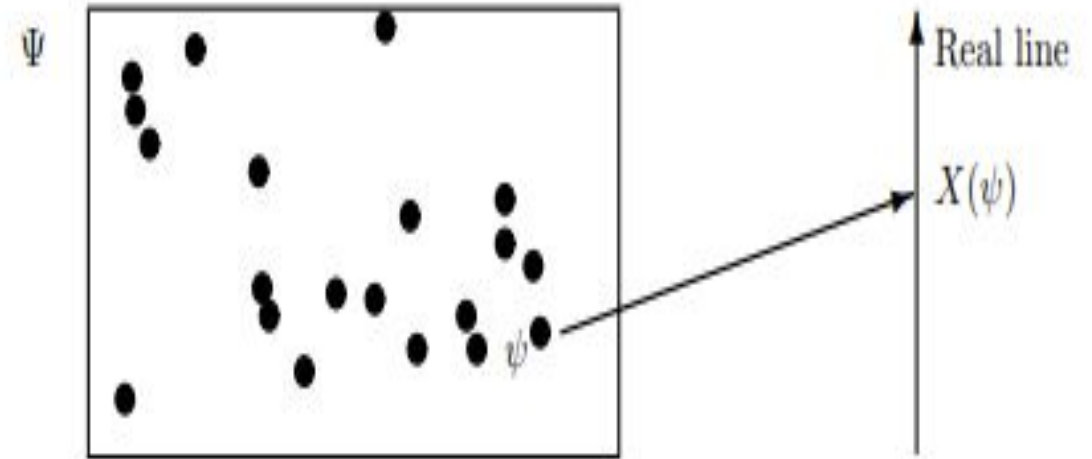
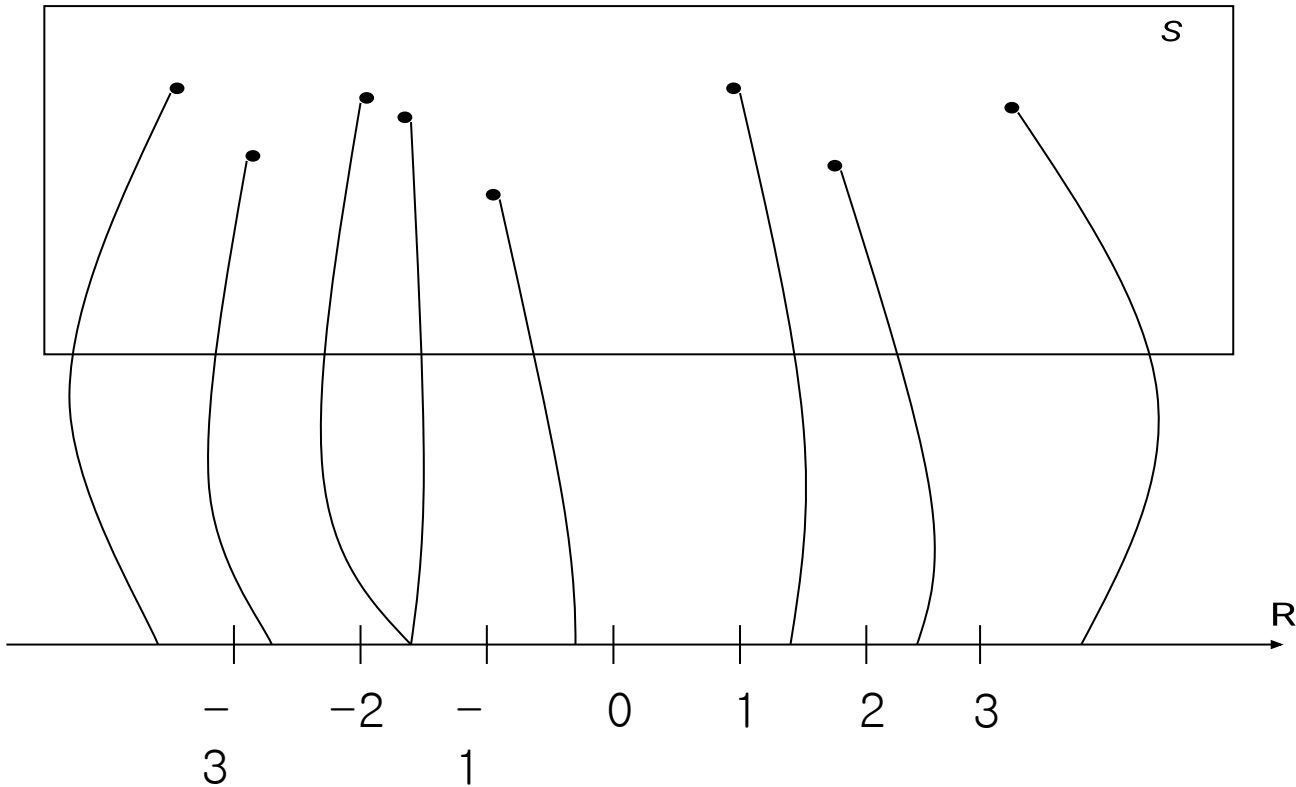
PSOT

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Definition of a Random Variable

- Random variable
 - A numerical value to each outcome of a particular experiment



Discrete Random Variable

- Consider our coin toss again. We could have heads or tails as possible outcomes.
- If we defined a variable, x , as the number of heads in a single toss, then x could possibly be 1 or 0, nothing else.
- Such a function, x , would be an example of a discrete random variable. **Such random variables can only take on discrete values.**
- Other examples would be the possible **results of a dice, or the number of students in a class room.**

Discrete Random Variable

- Example 1 : Machine Breakdowns
 - Sample space : $S = \{electrical, mechanical, misuse\}$
 - Each of these failures may be associated with a repair cost
 - State space : $\{50, 200, 350\}$
 - Cost is a random variable : 50, 200, and 350

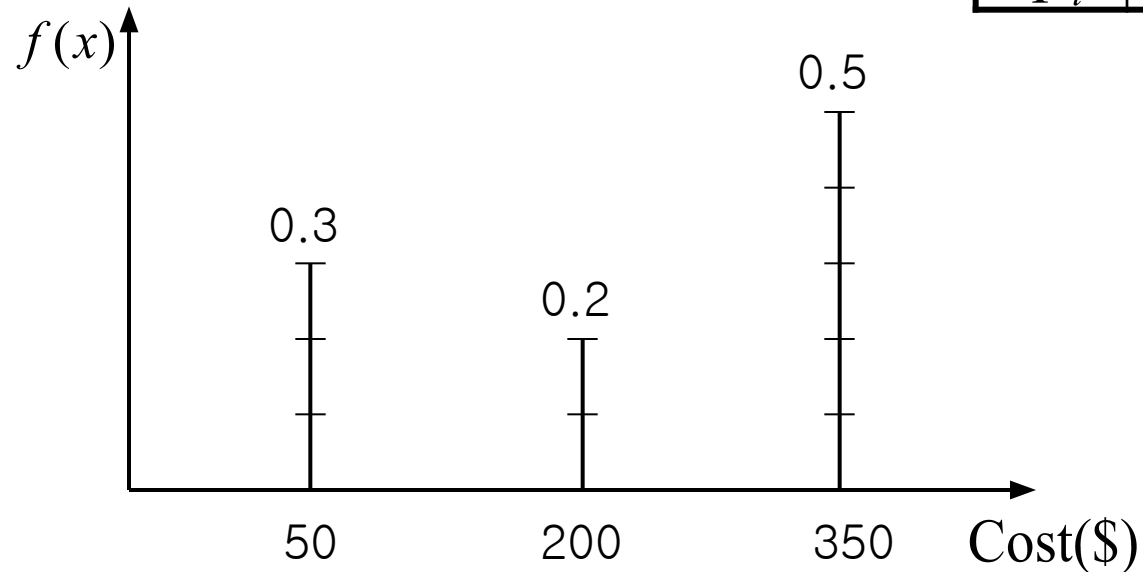
Probability Mass Function

- Probability Mass Function (p.m.f.)
 - A set of probability value p_i assigned to each of the values taken by the discrete random x_i variable
 - $0 \leq p_i \leq 1$ and $\sum_i p_i = 1$
 - Probability : $P(X = x_i) = p_i$

Probability Mass Function

- Example 1 : Machine Breakdowns
 - $P(\text{cost}=50)=0.3$, $P(\text{cost}=200)=0.2$,
 $P(\text{cost}=350)=0.5$
 - $0.3 + 0.2 + 0.5 = 1$

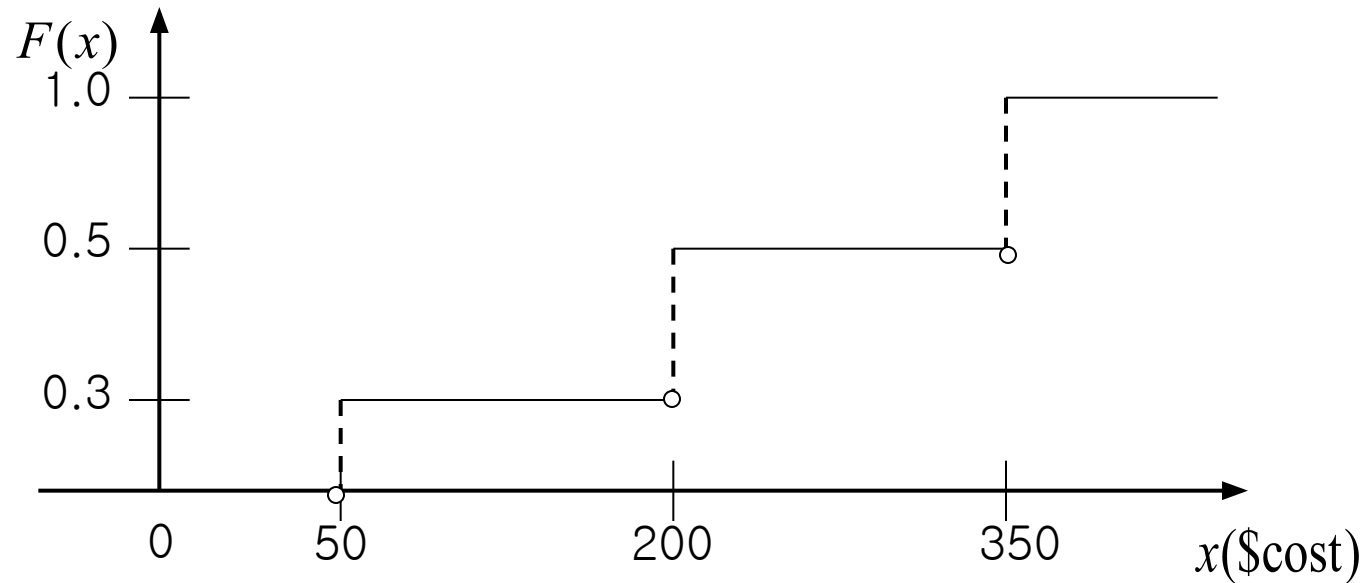
x_i	50	200	350
p_i	0.3	0.2	0.5



Cumulative Distribution Function

- Cumulative Distribution Function

- Function : $F(x) = P(X \leq x)$ $F(x) = \sum_{y: y \leq x} P(X = y)$
- Abbreviation : c.d.f



Cumulative Distribution Function

- Example 1 : Machine Breakdowns

$$-\infty < x < 50 \Rightarrow F(x) = P(\text{cost} \leq x) = 0$$

$$50 \leq x < 200 \Rightarrow F(x) = P(\text{cost} \leq x) = 0.3$$

$$200 \leq x < 350 \Rightarrow F(x) = P(\text{cost} \leq x) = 0.3 + 0.2 = 0.5$$

$$350 \leq x < \infty \Rightarrow F(x) = P(\text{cost} \leq x) = 0.3 + 0.2 + 0.5 = 1.0$$

Continuous Random Variables

- Back to the coin toss, what if we wished to **describe the distance between where our coin came to rest and where it first hit the ground.**
- That distance, x , would be a continuous random variable because **it could take on a infinite number of values within the continuous range of real numbers.** The coin could travel 1 cm, or 1.1 cm, or 1.11 cm, or on and on.
- Other examples of continuous random variables would be Time to Complete a Task, Voltage in an Electrical Circuit, Temperature of a Device, Weight of a Manufactured Product, Noise Level in a Signal

Continuous Random Variables

- Example : Metal Cylinder Production
 - Suppose that the random variable X is the diameter of a randomly chosen cylinder manufactured by the company. Since this random variable can take **any value between 49.5 and 50.5**, it is a **continuous** random variable.

Probability Density Function

- Probability Density Function (p.d.f.)
 - Probabilistic properties of a continuous random variable

$$f(x) \geq 0$$

$$\int_{\text{statespace}} f(x) dx = 1$$

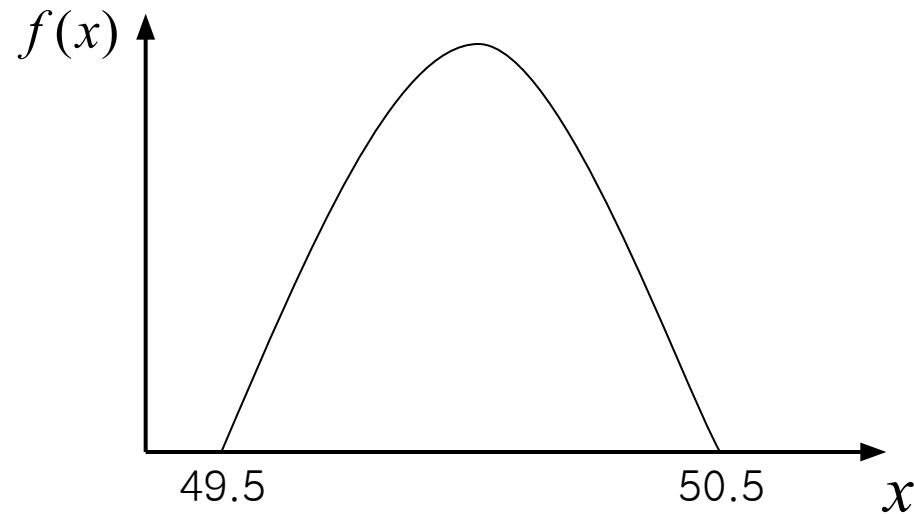
Probability Density Function

- Example

- Suppose that the diameter of a metal cylinder has a p.d.f

$$f(x) = 1.5 - 6(x - 50.2)^2 \quad \text{for } 49.5 \leq x \leq 50.5$$

$$f(x) = 0, \quad \text{elsewhere}$$



Probability Density Function

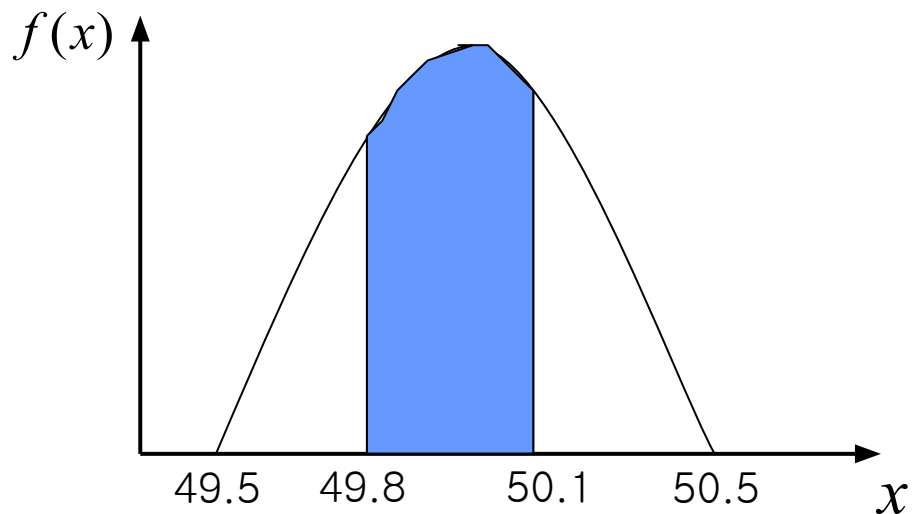
- This is a valid p.d.f.

$$\begin{aligned}\int_{49.5}^{50.5} (1.5 - 6(x - 50.0)^2) dx &= [1.5x - 2(x - 50.0)^3]_{49.5}^{50.5} \\ &= [1.5 \times 50.5 - 2(50.5 - 50.0)^3] \\ &\quad - [1.5 \times 49.5 - 2(49.5 - 50.0)^3] \\ &= 75.5 - 74.5 = 1.0\end{aligned}$$

Probability Density Function

- The probability that a metal cylinder has a diameter between 49.8 and 50.1 mm can be calculated to be

$$\begin{aligned}\int_{49.8}^{50.1} (1.5 - 6(x - 50.0)^2) dx &= [1.5x - 2(x - 50.0)^3]_{49.8}^{50.1} \\ &= [1.5 \times 50.1 - 2(50.1 - 50.0)^3] \\ &\quad - [1.5 \times 49.8 - 2(49.8 - 50.0)^3] \\ &= 75.148 - 74.716 = 0.432\end{aligned}$$



Cumulative Distribution Function

- Cumulative Distribution Function

- $F(x) = P(X \leq x) = \int_{-\infty}^x f(y)dy$

- $f(x) = \frac{dF(x)}{dx}$

- $P(a < X \leq b) = P(X \leq b) - P(X \leq a)$
 $= F(b) - F(a)$

- $P(a \leq X \leq b) = P(a < X \leq b)$

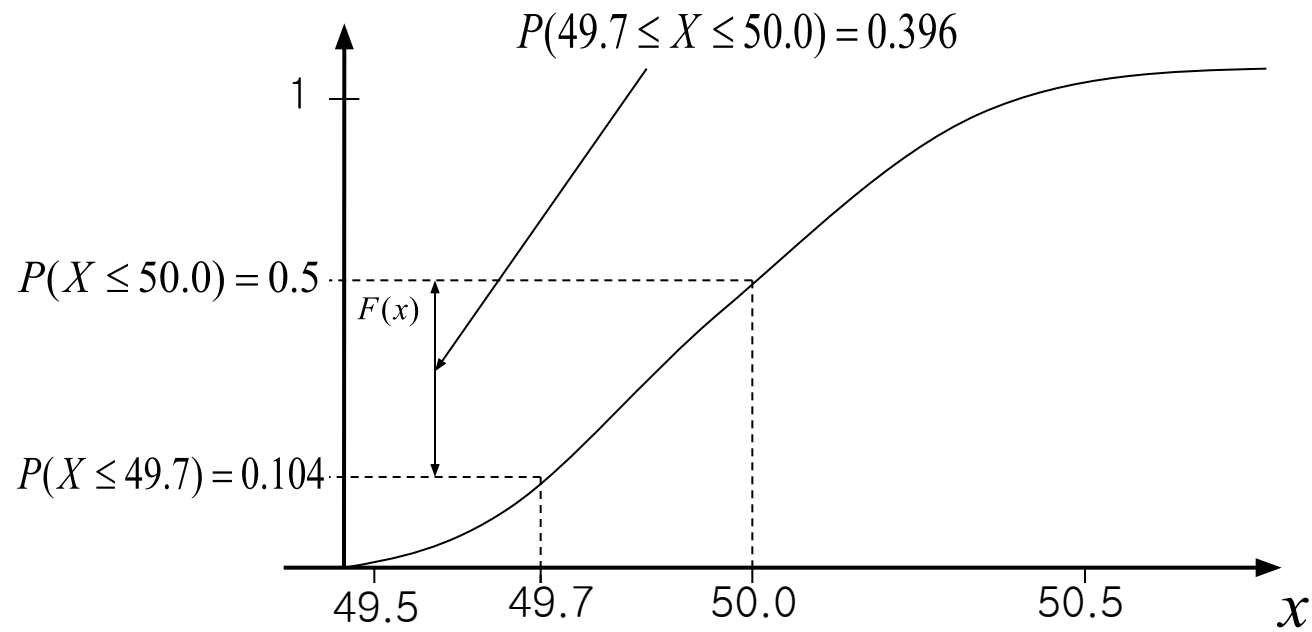
Probability Density Function

- Example

$$\begin{aligned} F(x) = P(X \leq x) &= \int_{49.5}^x (1.5 - 6(y - 50.0)^2) dy \\ &= [1.5y - 2(y - 50.0)^3]_{49.5}^x \\ &= [1.5x - 2(x - 50.0)^3] - [1.5 \times 49.5 - 2(49.5 - 50.0)^3] \\ &= 1.5x - 2(x - 50.0)^3 - 74.5 \end{aligned}$$

$$\begin{aligned} P(49.7 \leq X \leq 50.0) &= F(50.0) - F(49.7) \\ &= (1.5 \times 50.0 - 2(50.0 - 50.0)^3 - 74.5) \\ &\quad - (1.5 \times 49.7 - 2(49.7 - 50.0)^3 - 74.5) \\ &= 0.5 - 0.104 = 0.396 \end{aligned}$$

Cumulative Distribution Function



If the probability distribution of a discrete random variable X is given as

$$P(X = x) = \begin{cases} \frac{x}{15}, & \text{for } x = 1, 2, 3, 4, 5 \\ 0, & \text{otherwise} \end{cases}$$

The probability distribution can be obtained as

$X = x$	1	2	3	4	5
$P(X = x)$	1/15	2/15	3/15	4/15	5/15

(Continuous random variable case) Diameter of an electric cable, say X , is assumed to be a continuous variable with the function:

$$f(x) = \begin{cases} 6x(1-x), & \text{for } 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Check whether $f(x)$ is a probability density function.

(i) From property (i) for PDF, it is obvious that for $0 \leq x \leq 1$, $f(x) \geq 0$. Now for property (ii) for PDF, consider

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 6x(1-x) dx$$

or

$$\int_0^1 6(x - x^2) dx = 6 \left| \frac{x^2}{2} - \frac{x^3}{3} \right|_0^1 = 1$$

Therefore, $f(x) = 6x(1-x)$, $0 \leq x \leq 1$ is a PDF of the random variable X .

Suppose that the error in the reaction temperature in °C, for a controlled laboratory experiment, is a continuous random variable X having the PDF

$$f(x) = \begin{cases} \frac{x^2}{3}, & \text{for } -1 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Obtain the cumulative function $F(x)$, and use the same to evaluate $P(0 < X < 1)$.

We know that by definition

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

or

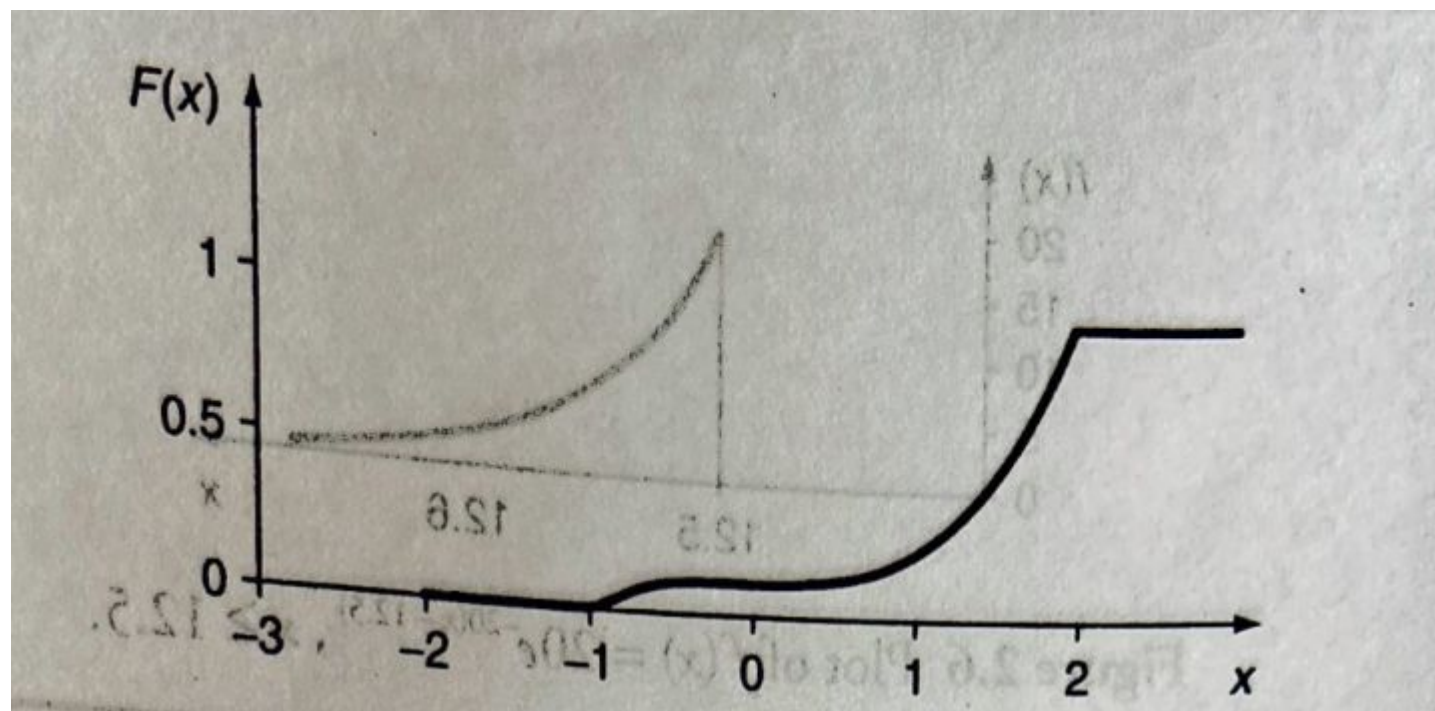
$$F(x) = \int_{-1}^x \frac{x^2}{3} dx = \left| \frac{x^3}{9} \right|_{-1}^2 = \frac{x^3 + 1}{9}$$

Therefore,

$$F(x) = \begin{cases} 0, & \text{when } x < -1 \\ \frac{x^3 + 1}{9}, & \text{when } -1 < x < 2 \\ 1, & \text{when } x > 2 \end{cases}$$

Now from the properties of cumulative distribution function $F(x)$, we have

$$P(0 < x < 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}$$



If x is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} kx, & \text{when } 0 < x < 2 \\ 2k, & \text{when } 2 < x < 4 \\ k(6-x), & \text{when } 4 < x < 6 \\ 0, & \text{otherwise} \end{cases}$$

Find the value of k and also find the cumulative distribution function $F(x)$. Also verify.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 k(6-x) dx = 1$$

$$\left| \frac{kx^2}{2} \right|_0^2 + \left| 2kx \right|_2^4 + \left| k \left(6x - \frac{x^2}{2} \right) \right|_4^6 = 1$$

On simplification, we get $k = 1/8$.

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

When $0 < x < 2$, we have

$$F(x) = \int_0^x \frac{x}{8} dx = \left| \frac{x^2}{16} \right|_0^x = \frac{x^2}{16}$$

When $2 < x < 4$, we have

$$F(x) = \int_0^2 \frac{x}{8} dx + \int_2^x \frac{2}{8} dx = \left| \frac{x^2}{16} \right|_0^2 + \left| \frac{2x}{8} \right|_2^x = \frac{x-1}{4}$$

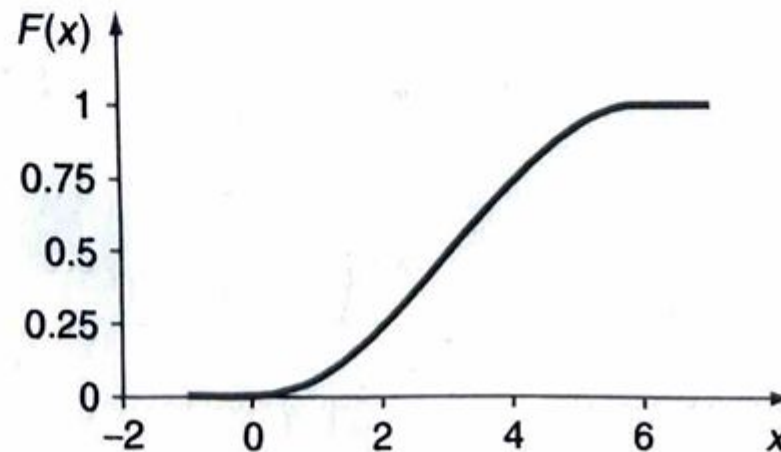
When $4 < x < 6$, we have

$$F(x) = \int_0^2 \frac{x}{8} dx + \int_2^4 \frac{2}{8} dx + \int_4^x \frac{1}{8} (6-x) dx = \left| \frac{x^2}{16} \right|_0^2 + \left| \frac{2x}{8} \right|_2^4 + \left| \frac{1}{8} \left(6x - \frac{x^2}{2} \right) \right|_4^x = \frac{12x - x^2 - 20}{16}$$

Therefore,

$$F(x) = \begin{cases} 0, & \text{when } x < 0 \\ \frac{x^2}{16}, & \text{when } 0 < x < 2 \\ \frac{x-1}{4}, & \text{when } 2 < x < 4 \\ \frac{12x - x^2 - 20}{16}, & \text{when } 4 < x < 6 \\ 1, & \text{when } x > 6 \end{cases}$$

A graphical representation to $F(x)$ is shown in Figure 2.8.



A random variable X takes the values $-1, 1, 3$ with equal probabilities and 5 with probability $1/2$. Then find (i) the probability distribution of X and (ii) $P(|X - 3| > 1)$.

$$P(X = -1) = P(X = 1) = P(X = 3) = k$$

$$3k + \frac{1}{2} = 1 \Rightarrow 3k = \frac{1}{2} \Rightarrow k = \frac{1}{6}$$

Therefore, the probability distribution is given as in the following table

$X = x$	-1	1	3	5
$P(X = x)$	$1/6$	$1/6$	$1/6$	$1/2$

$$\begin{aligned}P(|X - 3| > 1) &= 1 - P(|X - 3| \leq 1) \\&= 1 - P(-1 \leq X - 3 \leq +1) \\&= 1 - P(2 \leq X \leq 4) \\&= 1 - \left(\frac{1}{6}\right) \\&= \frac{5}{6}\end{aligned}$$

